Day 121 | Unit 09 | Teacher plan

Display, Describe, and Summarize Data

Recognize Statistical Questions

Standards: 6.SP.A.1

Learning target: I can recognize a statistical question and the expected variability.

enVision Topic 8: Display, Describe, and Summarize Data | Spectrum companion Chapter 7

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. A statistical question anticipates different answers in a collection of data. Name the group, the measured attribute, and the expected variation. A question about one fixed fact is not statistical.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references:

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01 / Learn

A statistical question anticipates different answers in a collection of data. Name the group, the measured attribute, and the expected variation. A question about one fixed fact is not statistical.

Worked example 1

Is “What are the wingspans, in centimeters, of the birds observed at a feeder?” statistical? Explain the group, attribute, units, and variation.

1. A statistical question anticipates different answers in a collection of data.

2. Name the group, the measured attribute, and the expected variation.

3. The question anticipates a collection of measurements.

4. Each bird can have a different wingspan.

5. Use the same unit for every observation.

6. State the result in the form requested: Yes; observed birds; wingspan; centimeters; wingspans can vary..

Answer: Yes; observed birds; wingspan; centimeters; wingspans can vary.

Worked example 2

A class studies a collection of seedlings. Select all statistical questions, and identify the expected variability.
A. How does height vary among the seedlings?
B. What is 6 + 2?
C. What is the height in centimeters of each of the seedlings?
D. How many centimeters are in 6 meters?

1. A statistical question anticipates different answers in a collection of data.

2. Different seedlings can have different height.

3. The questions anticipate variation across a collection.

4. The conversion and arithmetic questions have fixed answers.

Answer: A, C - How does height vary among the seedlings?; What is the height in centimeters of each of the seedlings?

Day 121 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Is “How many pages did each student read last night?” statistical? Explain.

02 / Constructed response

Is “How many centimeters are in one meter?” statistical? Explain.

03 / Constructed response

Is “What are the heights of the plants in our garden?” statistical? Explain.

04 / Constructed response

Identify the group, attribute, and units in: How many minutes do students in our class sleep each night?

05 / Constructed response

Rewrite “How long is my pencil?” as a statistical question.

06 / Constructed response

A student asks only one person’s age once. Explain why “How old is this person today?” is not statistical.

07 / Constructed response

Write a statistical question about travel to school. Name the group, attribute, units, and why answers may vary.

08 / Multiple select

A class studies a collection of students. Select all statistical questions, and identify the expected variability.
A. How many centimeters are in 4 meters?
B. What is the reading time in minutes of each of the students?
C. What is 4 + 5?
D. How does reading time vary among the students?

09 / Multiple select

A class studies a collection of backpacks. Select all statistical questions, and identify the expected variability.
A. How does mass vary among the backpacks?
B. How many centimeters are in 3 meters?
C. What is the mass in kilograms of each of the backpacks?
D. What is 3 + 4?

10 / Constructed response

Critique: “A question is statistical whenever it contains numbers.” Give a counterexample and a corrected rule.

Day 121 | Unit 09 | Answer key

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1. Yes; pages read vary across students.

Look for a collection of data and anticipated variation; identify the population and variable.

6.SP.A.1 | Constructed response

2. No; one fixed conversion, 100 cm.

Look for a collection of data and anticipated variation; identify the population and variable.

6.SP.A.1 | Constructed response

3. Yes; plant heights vary.

Look for a collection of data and anticipated variation; identify the population and variable.

6.SP.A.1 | Constructed response

4. Group: class; attribute: sleep duration; units: minutes.

Look for a collection of data and anticipated variation; identify the population and variable.

6.SP.A.1 | Constructed response

5. Example: What are the lengths in centimeters of the pencils in our classroom?

Look for a collection of data and anticipated variation; identify the population and variable.

6.SP.A.1 | Constructed response

6. It seeks one fixed value, rather than anticipating variation in a data collection.

Look for a collection of data and anticipated variation; identify the population and variable.

6.SP.A.1 | Constructed response

7. Example: How many minutes do students in our class take to reach school? Group: class; travel time in minutes varies with routes and travel methods.

Look for a collection of data and anticipated variation; identify the population and variable.

6.SP.A.1 | Constructed response

8. B, D - What is the reading time in minutes of each of the students?; How does reading time vary among the students?

Different students can have different reading time. The questions anticipate variation across a collection. The conversion and arithmetic questions have fixed answers.

6.SP.A.1 | Multiple select

9. A, C - How does mass vary among the backpacks?; What is the mass in kilograms of each of the backpacks?

Different backpacks can have different mass. The questions anticipate variation across a collection. The conversion and arithmetic questions have fixed answers.

6.SP.A.1 | Multiple select

10. Counterexample: How many sides does a triangle have? It has a fixed answer. A statistical question anticipates variability in a collection of data.

Look for a collection of data and anticipated variation; identify the population and variable.

6.SP.A.1 | Constructed response